

Tafila Technical University
Faculty of Information Technology and Communication
Department of Computer Science

Course Information

Course Title: AI Programming

Course Code: 0602221

Homework Number: HW 4

Title: Roulette Game

Main topics

- Loop statements
- Built-in functions
- User-defined functions

Program Specification:

Write a **Python program** that allows the user to play **Rock–Paper–Scissors** against the **computer**. In this version of the game, the following rules apply:

- Rock beats Scissors
- Scissors beats Paper
- Paper beats Rock
- If both choices are the same → it is a tie

The user starts with **100 chips**, and each round the user plays against the **computer**.

Your program should behave as follows:

1. Display a welcome message
2. Repeatedly do the following:
 - Ask the user to choose to play or cash out
 - If the user chooses to play:
 - Ask the user to choose **Rock, Paper, or Scissors**
 - Ask the user to choose an amount to bet (between 1 chip and the total number of chips the user currently has), and record the bet amount

- The **computer randomly selects** Rock, Paper, or Scissors
 - Display both the user's choice and the computer's choice
 - Inform the user whether they **win, lose, or tie**
 - Update the user's chip total:
 - Win → add bet amount
 - Lose → subtract bet amount
 - Tie → no change
3. Continue until:
- a. The user chooses to cash out, or
 - b. The user has no more chips left

Final Report:

Once the user chooses to cash out, display the total **winnings or losses** compared to the starting 100 chips.

Rules and Requirements:

- All user input must be validated
- The game must be played **against the computer (not another user)**

Required Functions

```
def welcome():  
def get_menu_choice():  
def get_move():  
def get_bet(chips_now):  
def determine_winner(user, computer):  
def report(chips_now):
```

Sample Run(s):

```
#####  
# WELCOME TO RPS GAME  #  
#####
```

You have 100 chips

1. Play Rock-Paper-Scissors

2. Cash out

Enter a choice [1-2]: 1

Enter your move [Rock, Paper, Scissors]: rock

Enter the number of chips to bet [1-100]: 10

Computer chose: Scissors

Congrats, you won!

You now have 110 chips